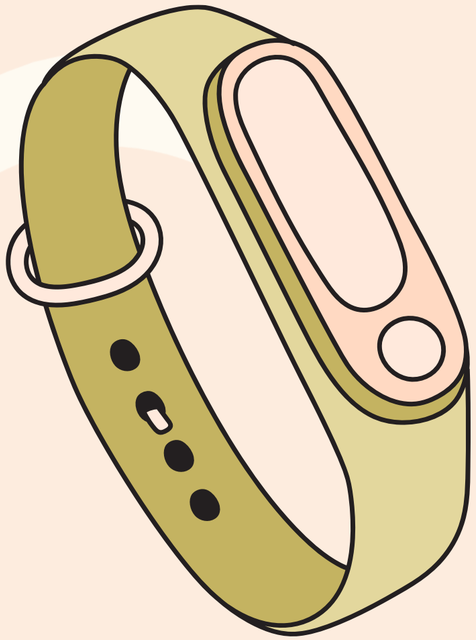
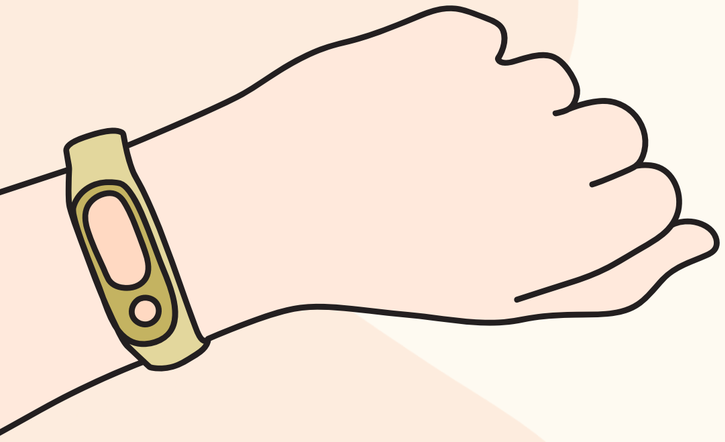
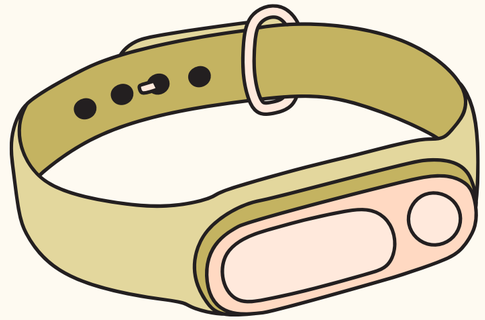
Smart Multi-Sensor Wearable for Real-time Elderly Health Anomaly and Fall Detection





Project ID	**R26-IT-108**
Research Group	**Autonomous Intelligent Machines and Systems (AIMS)**
Specialization	**Information Technology**
Supervisor	**Nelum Chathuranga Amarasena**

OUR TEAM

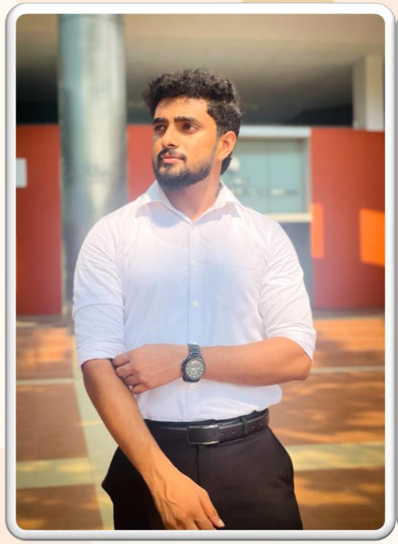




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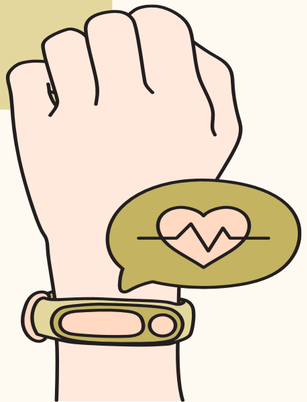


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IT.No : IT22146410

The Problem: Elderly Health Risks at Home

Elderly individuals living independently often experience undetected health emergencies such as:

- ❖ Falls
- ❖ Silent hypoxia (low oxygen levels without symptoms)
- ❖ Cardiac arrhythmias
- ❖ Stress or internal pain



- ➢ These conditions can develop without immediate warning, leading to delayed medical intervention.

Evidence

- ✓ Falls are the leading cause of injury-related deaths among elderly people.
- ✓ Silent physiological deterioration can occur hours before symptoms appear.

Users and Stakeholders

Primary Users

- individuals aged **65+ living independently**
- Seniors with **chronic health conditions**

Stakeholders

- Family members and caregivers
- Hospitals and emergency services
- Assisted living facilities
- Healthcare providers

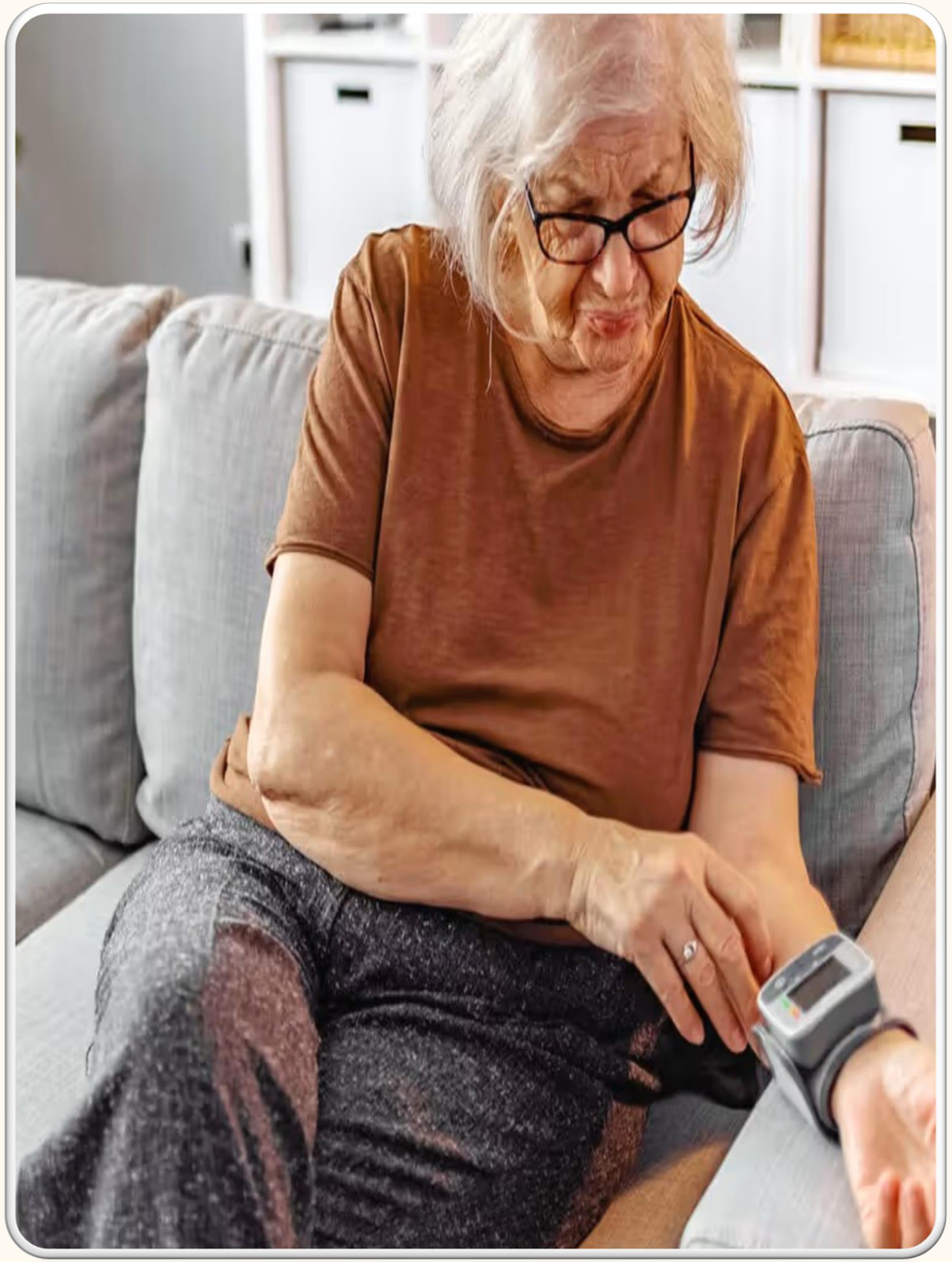
Impact of the Problem

Impact on Individuals

- Increased risk of injury and mortality
- Loss of independence
- Delayed emergency response

Impact on Society

- Higher healthcare costs
- Increased pressure on emergency services
- Emotional stress for families and caregivers

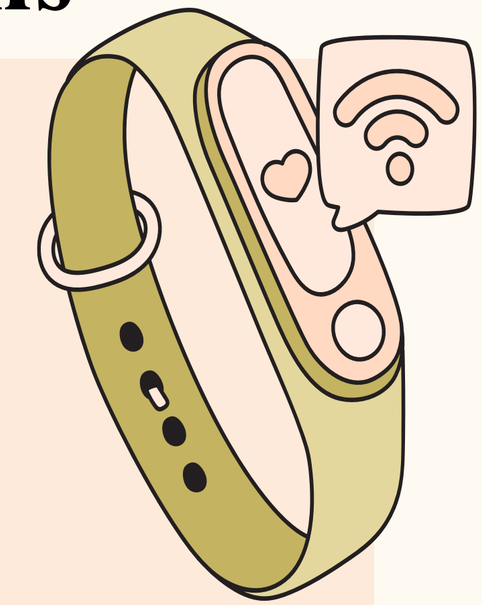


Existing Solutions and Limitations

Existing Systems

Examples:

- ✓ Apple Watch
- ✓ Fitbit
- ✓ Basic fall detectors



Limitations

- ❑ High false alarm rates
- ❑ Cloud dependency for analysis
- ❑ Limited health parameter monitoring
- ❑ Lack of emotional or neurological monitoring
- ❑ High cost for elderly users

Proposed Solution

Smart Multi-Sensor Health Monitoring Wearable

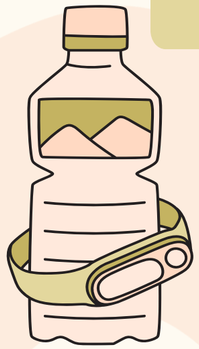
A wearable system that monitors multiple physiological signals and detects health anomalies in **real time**.

Key Features

- ❖ Multi-sensor data collection
- ❖ Edge-based AI processing
- ❖ Real-time anomaly detection
- ❖ Immediate alert generation

Sensors Used

- ❖ Heart Rate / SpO_2 Sensor (MAX30102)
- ❖ Temperature Sensor (MLX90614)
- ❖ Motion Sensor (MPU6050)
- ❖ GSR Sensor (Stress Detection)

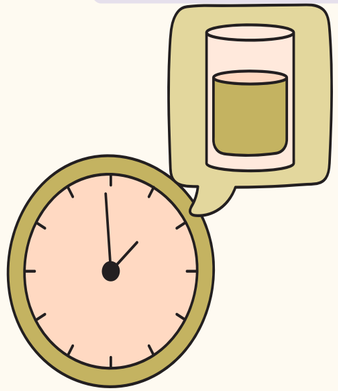


System Overview

Input Layer (Sensors)

Collect physiological and motion data:

- Heart rate
- Oxygen levels
- Skin temperature
- Motion patterns
- Skin conductance



Processing Layer

- ESP32 Microcontroller
- TinyML algorithms
- Sensor fusion analysis

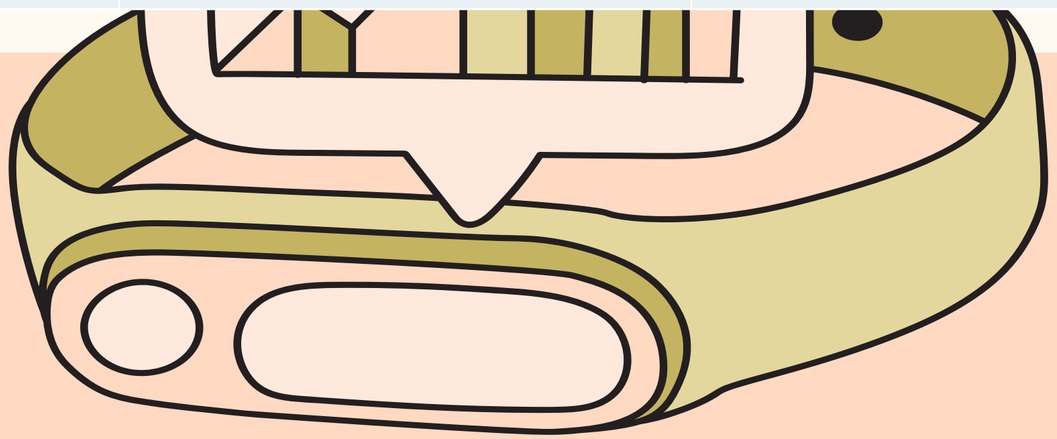
Output Layer

- Local vibration alerts
- Mobile application notifications
- Emergency alert system



Individual Components (Team Contribution)

Module	Researcher	Function
Cardiovascular Anomaly Detection	Abeysinghe	Detect abnormal heart rhythms
Kinematic Fall Detection	Lakshan	Identify falls and immobility
Respiratory & Thermal Warning	Dissanayaka	Detect hypoxia and sepsis
Neurological Stress Monitoring	Peshala	Detect emotional stress and pain



Cardiovascular Anomaly Detection System

IT22081148 | Abeysinghe S.A.M.M.

Problem Definition

- Cardiovascular diseases are one of the leading causes of death among elderly individuals. Many wearable devices measure heart rate but cannot detect **irregular heart rhythms (arrhythmia)** in real time.
  In many existing systems, data is sent to cloud servers for analysis, which introduces **latency, privacy risks, and dependency on internet connectivity**.

Research Objective

- To design and develop a **privacy-centric cardiovascular anomaly detection system** that uses Edge AI to detect abnormal heart rhythms directly on a wearable device.

Specific Objectives

- Acquire real-time PPG signals using the **MAX30102 sensor**
- Develop a **signal preprocessing pipeline** to remove noise
- Train a **1D Convolutional Neural Network (1D-CNN)** using the MIT-BIH Arrhythmia dataset
- Deploy the optimized model on the **ESP32 microcontroller** using TinyML

Novelty

- Real-time arrhythmia detection on a **low-power wearable device**
- **Edge AI processing without cloud dependency**
- Improved **data privacy and faster emergency response**

Research Methodology & Domain Knowledge

Technologies and Tools

Hardware

- ESP32 microcontroller
- MAX30102 PPG sensor

Software

- Python with TensorFlow for model training
- TensorFlow Lite Micro for embedded inference
- PlatformIO for firmware development

Methodology

1. **Signal Acquisition**
    - PPG signals collected from MAX30102 sensor
2. **Signal Processing**
    - Noise filtering
    - Motion artifact removal
    - Feature extraction
3. **Machine Learning Model**
    - 1D-CNN trained using **MIT-BIH Arrhythmia dataset**
4. **Edge Deployment**
    - Model quantization for ESP32
    - Real-time arrhythmia classification

Validation

- Dataset: MIT-BIH Arrhythmia Database
- Performance metrics: Accuracy, Precision, Recall, and F1-score
- Real-time testing with wearable sensor signals

System Overview

❑ The cardiovascular monitoring module follows the pipeline:

PPG Sensor → Signal Processing → 1D-CNN Model → Heart Rhythm Classification

Output

❑ The system classifies heart rhythm into:

- Normal heartbeat
- Abnormal cardiac anomaly

❑ If an anomaly is detected:

- The wearable device generates a **vibration alert**
- A warning notification is sent to the **mobile application**

Dependencies & Integration

❑ This module works with other system components:

- **Fall Detection Module** – correlates heart rate with movement
- **Respiratory Monitoring Module** – combines HR with oxygen levels
- **Stress Monitoring Module** – analyzes HR with stress signals

Through **sensor fusion**, this module contributes to the overall **multi-sensor intelligent health monitoring system**.

Kinematic Fall & Immobility Detection System

IT22196156 | Lakshan N.K.T.

Problem Definition

- Falls are one of the **leading causes of injury and hospitalization among elderly individuals**. Many wearable fall detection systems generate **false alarms** during daily activities such as sitting quickly, clapping, or dropping objects.
- Frequent false alarms cause **alarm fatigue**, leading users to disable the device and reducing its effectiveness.

Research Objective

- To design and implement an **accurate fall detection system** using motion sensors and edge-based algorithms that can distinguish real falls from normal daily activities.

Specific Objectives

- Integrate **MPU6050 6-axis IMU sensor** with ESP32
- Capture real-time motion data at high frequency
- Develop a **two-stage fall detection algorithm**
- Implement a lightweight **decision-tree model** for on-device detection
- Reduce false positives compared to traditional threshold systems

Novelty

- **Two-stage verification fall detection system**
- Edge-based decision logic running on ESP32
- Improved accuracy and **reduced false alarms**
- Real-time detection without cloud processing

Research Methodology & Domain Knowledge

Technologies and Tools

Hardware

- ESP32 microcontroller
- MPU6050 Inertial Measurement Unit (IMU)

Software

- Arduino Framework / PlatformIO
- Python for dataset processing
- C++ for embedded algorithm implementation

Methodology

1. Motion Data Acquisition

- 6-axis motion data collected from accelerometer and gyroscope

2. Feature Extraction

- Resultant Vector Magnitude (RVM)
- Signal Magnitude Area (SMA)
- Orientation angles (pitch and roll)

3. Two-Stage Fall Detection Algorithm

Stage 1 – Impact Detection

- Identify sudden acceleration spikes

Stage 2 – Orientation Verification

- Check if user remains in horizontal position after impact

Validation

- Benchmark datasets: **SisFall and Kfall**
- Performance evaluation using sensitivity and false alarm rate

System Workflow

IMU Motion Sensor → Motion Feature Extraction → Decision Tree Algorithm → Fall Detection

Detection Process

1. Detect **high-impact acceleration spike**
2. Monitor **body orientation and immobility**
3. Confirm fall event before triggering alert

Output

If a fall is detected:

- Wearable device triggers **vibration alert**
- Emergency notification sent to **mobile application**
- Caregivers receive real-time alert

Through **sensor fusion**, this module contributes to the overall **multi-sensor intelligent health monitoring system**.

Integration with Full System

This module provides **motion context** for the entire wearable system.

Examples of integration:

- Fall detected + abnormal heart rate → emergency situation
- High motion + high heart rate → normal physical activity
- Fall + immobility → critical alert

This module improves the reliability of the **multi-sensor intelligent health monitoring system**.

Respiratory & Thermal Warning System

IT22291196 | Dissanayaka D.M.S.K.

Problem Definition

- Elderly individuals often experience serious health conditions such as **silent hypoxia and sepsis**, where vital signs deteriorate before symptoms become noticeable.
- Traditional monitoring systems rely on **static threshold values** (for example SpO_2 below 90%), which can cause frequent false alarms or miss early warning signs.
- This delay between physiological deterioration and symptom awareness creates a **diagnostic gap** in home healthcare monitoring.

Research Objective

- To develop a **smart respiratory and thermal monitoring system** capable of detecting early signs of physiological deterioration using multi-sensor data analysis.

Specific Objectives

- Integrate **MAX30102 pulse oximeter** to monitor SpO_2 and heart rate
- Integrate **MLX90614 infrared sensor** to measure body temperature
- Develop a **multi-parameter anomaly detection algorithm**
- Implement **K-Means clustering** for personalized health baseline detection
- Deploy the algorithm on the **ESP32 microcontroller for real-time monitoring**

Novelty

- **Multi-parameter vital sign correlation** instead of single-threshold detection
- **Personalized physiological baseline (Homeostasis Zone)**
- Edge-based anomaly detection using **unsupervised machine learning**

Research Methodology & Domain Knowledge

Technologies Used

Hardware

- ESP32 microcontroller
- MAX30102 pulse oximeter sensor
- MLX90614 infrared temperature sensor

Software

- Python (Scikit-learn) for model development
- Arduino / C++ for embedded system implementation

Methodology

1. **Data Acquisition**
   - Collect physiological signals including SpO_2, heart rate, and skin temperature
2. **Signal Preprocessing**
   - Noise filtering
   - Sensor calibration

3. Unsupervised Machine Learning

- Use **K-Means clustering** to establish a personalized vital sign baseline

4. Anomaly Detection

- Measure deviation from baseline using Euclidean distance

Validation

➢ Clinical dataset reference: **MIMIC-III dataset**

➢ Performance evaluated using **accuracy and anomaly detection rate**

System Workflow

Temperature Sensor + SpO_2 Sensor → Data Processing → K-Means Algorithm → Health Status Classification

Detection Capabilities

The systemcan detect early warning signs of :

- **Silent hypoxia** (low oxygen levels)
- **Early-stage infection or sepsis**
- **Respiratory deterioration**

Alert Mechanism

When abnormal patterns are detected:

- Wearable device triggers **vibration alert**
- Notification sent to **mobile caregiver application**

Integration with Full System

This module contributes **vital sign monitoring** to the overall wearable system.

Examples of integration:

- Temperature increase + elevated heart rate → infection risk
- Low SpO_2 + immobility → respiratory emergency
- Stress signals + abnormal vitals → potential health deterioration

This module enhances the system's ability to perform **predictive health monitoring for elderly individuals**.

Neurological Stress & Pain Monitoring System

IT22146410 | Peshala E.A.P.

Problem Definition

- Many elderly patients, especially those with **dementia, Alzheimer's disease, or stroke**, may lose the ability to verbally communicate their pain or emotional distress. As a result, caregivers often rely on **subjective observation**, which can lead to delayed treatment and reduced quality of care.

- Existing wearable devices mainly monitor physical parameters such as heart rate and step count but **do not detect emotional stress or internal pain conditions**.

Research Objective

- To design and implement a **Neurological Stress and Pain Monitoring system** that can detect emotional distress in elderly individuals using physiological signals.

Specific Objectives

- Measure **Galvanic Skin Response (GSR)** to detect emotional arousal
- Analyze Electrodermal Activity (EDA) signals
- Train a **Random Forest machine learning model** using the WESAD dataset
- Deploy the optimized model on the **ESP32 microcontroller**
- Detect stress and pain events in real time

Novelty

- Introduction of **Affective Computing in elderly healthcare wearables**
- Objective detection of **stress and pain using physiological signals**
- Edge-based machine learning processing for **real-time emotional state detection**

Technologies Used

Hardware

- ESP32 microcontroller
- Galvanic Skin Response (GSR) sensor
- Skin conductance electrodes

Software

- Python (Scikit-Learn) for machine learning model training
- PlatformIO / Arduino framework for embedded implementation

Methodology

1. Physiological Signal Acquisition

- Continuous monitoring of skin conductance using GSR sensor

2. Signal Processing

- Separation of tonic and phasic Electrodermal Activity (EDA) components
- Detection of Skin Conductance Response (SCR) peaks

3. Machine Learning Classification

- Random Forest model trained using **WESAD dataset**

4. Edge Deployment

- Model optimized and deployed on ESP32 for real-time detection

Validation

➢ Dataset used: **WESAD – Wearable Stress and Affect Detection Dataset**

➢ Evaluation metrics: Accuracy, Precision, and F1-score

System Workflow

GSR Sensor → Signal Processing → Feature Extraction → Random Forest Model → Stress Classification

Detection Capabilities

The system identifies physiological indicators of:

- **Silent hypoxia** (low oxygen levels)
- **Early-stage infection or sepsis**
- **Respiratory deterioration**

➢ This module enhances the wearable system by providing **psychological and neurological monitoring**, enabling a **more holistic approach to elderly healthcare monitoring**.

Alert Mechanism

If a distress state is detected:

- Wearable device triggers **vibration alert**
- Notification sent to **mobile caregiver application**

Integration with Full System

This module provides **emotional state monitoring** within the multi-sensor wearable system.

Examples of integration:

- Stress signal + high heart rate → emotional distress
- Stress signal + low motion → pain condition
- Stress signal + abnormal vital signs → medical alert

Commercialization

Market Opportunity

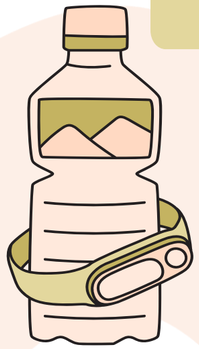
Growing demand for elderly healthcare monitoring devices.

Target Market

- ❖ Elderly individuals living alone
- ❖ Assisted living facilities
- ❖ Home healthcare services

Value Proposition

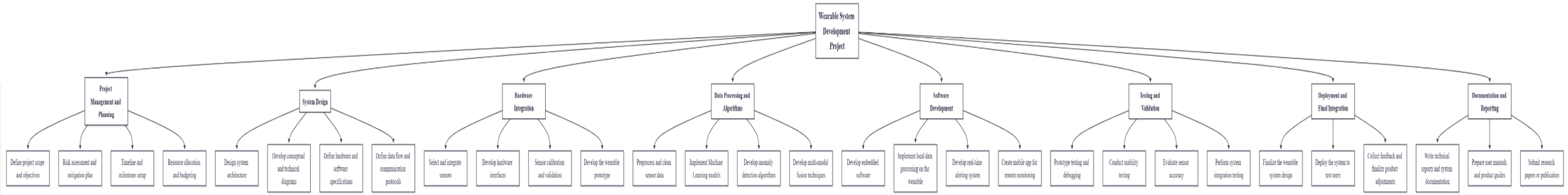
- ❖ Real-time health monitoring
- ❖ Privacy-preserving edge AI
- ❖ Multi-sensor health detection



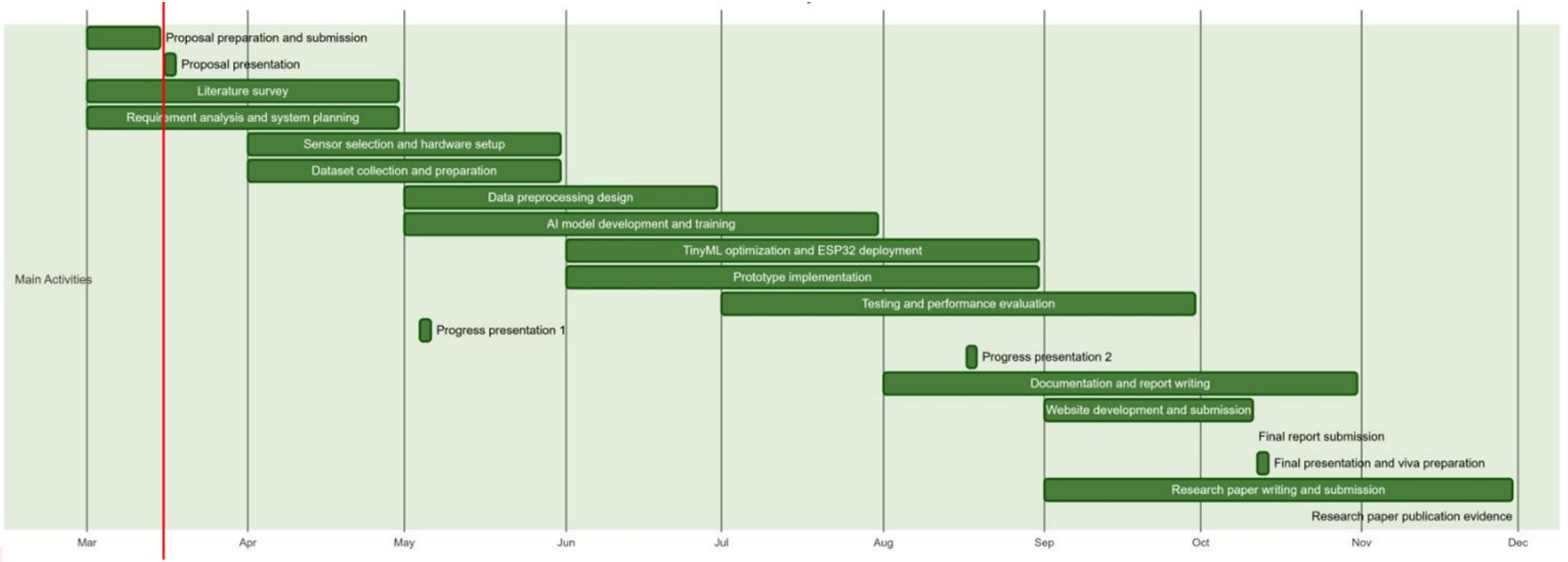
Project Planning

Work Breakdown Structure

Task	Subtasks
Project Management and Planning	Define scope, Risk assessment, Timeline setup
System Design	Design architecture, Define specifications, Data flow
Hardware Integration	Select sensors, Interface hardware, Develop prototype
Data Processing and Algorithms	Preprocess data, Implement ML models, Anomaly detection
Software Development	Develop embedded software, Mobile app, Alert system
Testing and Validation	Prototype testing, Usability testing, System integration
Deployment and Final Integration	Finalize design, Deploy to users, Collect feedback
Documentation and Reporting	Write reports, User manuals, Submit publications



Gantt Chart


Proposal preparation and submission
Proposal presentation
Literature survey
Requirement analysis and system planning
Sensor selection and hardware setup
Dataset collection and preparation
Data preprocessing design
AI model development and training
TinyML optimization and ESP32 deployment
Prototype implementation
Testing and performance evaluation
Progress presentation 1
Progress presentation 2
Documentation and report writing
Website development and submission
Final report submission
Final presentation and viva preparation
Research paper writing and submission
Research paper publication evidence
Main Activities
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec

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